

The Price of Carbon: Assessing the Impact of Carbon Taxation on Consumer Prices in South Africa

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Abstract

This paper examines the causal impact of South Africa's carbon tax on consumer prices using a quasi-experimental design based on the Generalized Synthetic Control Method (GSCM). By constructing a synthetic counterfactual from a pool of developing countries without carbon pricing, we assess whether the introduction of the tax in 2019 generated inflationary effects. The results show no statistically significant impact on headline consumer prices, indicating that the carbon tax has not been inflationary in aggregate. However, we find a small but persistent cumulative increase—about 1%—in energy prices relative to the counterfactual, with no spillovers to core or food prices. These findings are robust to alternative estimators and donor pool restrictions. Overall, the results suggest that South Africa's carbon tax primarily altered relative prices rather than aggregate inflation, reflecting the moderating influence of generous emissions allowances and phased implementation of the policy.

Keywords: Carbon tax; Inflation; Consumer prices; Greenflation; Synthetic control; South Africa.

JEL classification codes: E31, H23, Q43, Q54, Q58

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1 Introduction

Carbon pricing—whether implemented through a carbon tax or an emissions trading scheme (ETS)—has emerged as a leading market-based instrument for reducing greenhouse gas (GHG) emissions and steering economies toward low-carbon development (Hepburn et al., 2020).¹ As of April 2023, there are 73 carbon taxes or ETSs in operation worldwide, encompassing 23% of global GHG emissions (World Bank, 2023). Notably, this includes a few developing nations like Colombia and South Africa, while others like Nigeria, Senegal, and Gabon are currently contemplating the adoption of an ETS or carbon tax (World Bank, 2023). The expanding adoption of carbon pricing reflects a global recognition that pricing carbon is central to achieving sustainable growth and net-zero targets.

While the environmental effectiveness of carbon pricing is well established, a growing strand of the literature highlights potential macroeconomic side effects, particularly the risk of inflationary spillovers, often termed "greenflation" (IMF, 2022; Schnabel, 2022; Konradt and Weder di Mauro, 2023; Konradt et al., 2024; Moessner, 2025). Such concerns are especially salient in emerging economies, where energy and carbon-intensive goods account for a larger share of household expenditure and production costs (García and Suárez-Varela, 2022). Carbon pricing can increase the cost of fossil-fuel-based energy and, by extension, the cost of production for energy-intensive goods and services (Andersson et al., 2020). On the supply side, higher input costs may raise consumer prices as firms pass on the additional burden, while on the demand side, higher energy prices can reduce real incomes, consumption, and investment, thereby dampening inflationary pressure (ICC, 2023). Technological progress and renewable-energy expansion may further offset cost increases by improving efficiency and reducing the weight of energy in household consumption baskets (Andersson et al., 2020). Consequently, the net price impact of carbon pricing depends on contextual factors—such as the tax rate, coverage, and emission-allowance

¹Carbon tax and ETS are classified as direct carbon pricing instruments since they are directly imposed on the amount of GHG emissions produced by a particular product or activity.

structure—and remains an empirical question.

Existing empirical evidence is mixed and largely confined to advanced economies. For instance, [Berthold et al. \(2023\)](#) find that carbon-pricing shocks increased inflation in the European Union, whereas [Konradt and Weder di Mauro \(2023\)](#) and [Konradt et al. \(2024\)](#) report no inflationary effects in the Euro Area. Similarly, [Moessner \(2025\)](#) estimates that a US\$ 10 per ton CO₂ increase raises energy-price inflation by 0.8 percentage points but headline inflation by only 0.08 points. These findings suggest that carbon pricing primarily alters relative rather than aggregate price levels. However, there remains a dearth of causal evidence from developing and emerging countries, where energy intensity is higher, markets are less competitive, and policy design often incorporates transitional exemptions.

South Africa offers a unique setting to address this gap. In June 2019, the country became the first in Africa to introduce a national carbon tax—an important milestone in its commitment to reach net-zero emissions by 2050. The tax covers major sources of GHG emissions, including fuel combustion, industrial processes, and fugitive emissions, and allows up to 95 percent in emission allowances depending on sector and activity (see [section 2](#) for further details). These generous provisions were intended to give firms and households time to adjust while limiting short-term economic disruption, making South Africa an ideal setting for evaluating the short- to medium-term inflationary effects of carbon taxation in an emerging-market context.

This paper assesses the causal impact of South Africa’s carbon tax on consumer prices using a quasi-experimental approach. Identifying this impact is challenging because the counterfactual—what would have happened to prices in the absence of the tax—is unobservable. To address this challenge, we employ the Generalized Synthetic Control Method (GSCM) developed by [Xu \(2017\)](#), an extension of the Synthetic Control Method (SCM) introduced by ([Abadie and Gardeazabal, 2003](#); [Abadie et al., 2010](#)). The GSCM constructs a data-driven synthetic control that reproduces pre-treatment outcomes while accounting for both time and country fixed effects and producing standard errors for valid inference.

We use a quasi-experimental approach because assessing the causal impact of carbon tax on consumer prices is far from trivial as one cannot observe what would have happened to consumer prices in the absence of the implementation of the tax (the counterfactual). We thus employ the Generalized Synthetic Control Method (GSCM) developed by Xu (2017), which is a generalization of the original SCM (Abadie and Gardeazabal, 2003; Abadie et al., 2010). The GSCM uses a data-driven approach to create a synthetic control that best reproduces the outcomes of the treated group in the pre-treatment period. An advantage of the GSCM over the original SCM in our context is that it allows us to account for both country and time fixed effects, in addition to controlling for other factors that could influence consumer prices. It also allows for classical inference because it provides uncertainty estimates such as standard errors and confidence intervals.

Our empirical results indicate that South Africa’s carbon tax has not caused a statistically significant increase in headline consumer prices over 16 quarters following its introduction. However, we find a modest cumulative increase of about 1 percent in energy prices relative to the counterfactual, with no spillover to core or food inflation. These findings suggest that the tax has influenced relative rather than aggregate prices—consistent with earlier studies for developed economies (Konradt and Weder di Mauro, 2023; Konradt et al., 2024). The muted price effects likely reflect the policy’s design, including its relatively low rate and extensive emissions allowances, which have cushioned short-term cost pressures.

This paper contributes to three intertwined strands of the literature.

First, it extends the empirical literature on tax incidence, cost pass-through, and the price effects of carbon taxation by providing a causal, ex-post analysis of a carbon-pricing policy in an emerging-economy context. Previous studies have documented substantial pass-through of energy and emissions costs in industrial settings (Ahn, 2025) and near-complete pass-through of emissions costs to electricity prices (Fabra and Reguant, 2014). However, evidence on whether carbon taxation leads to broad consumer price inflation remains mixed (Berthold et al., 2023; Konradt and Weder di

Mauro, 2023; Konradt et al., 2024; Moessner, 2025).

Second, the paper contributes to the literature on carbon-tax design and its macroeconomic consequences in emerging economies. While much of the existing research focuses on developed countries, this study complements and contrasts with earlier ex-ante assessments of South Africa's carbon tax that examined its macroeconomic (Alton et al., 2012; Nong, 2020) and distributional (Okonkwo, 2021; Dorband et al., 2019) effects. By applying the Generalized Synthetic Control Method (GSCM) to the South African case, the paper provides rigorous ex-post, quasi-experimental evidence on how a comprehensive carbon tax affects headline, core, food, and energy consumer prices.

Third, the paper advances the methodological and policy debate on greenflation and the macroeconomic risks of climate policy. Through the application of the GSCM, it contributes to the growing empirical toolkit for evaluating environmental policies in data-limited contexts and demonstrates that the inflationary impact of South Africa's carbon tax is largely confined to direct energy-related channels, with no evidence of broader inflationary spillovers. This finding supports the view that carbon pricing primarily alters relative rather than aggregate price levels (Konradt and Weder di Mauro, 2023; Konradt et al., 2024; Moessner, 2025). From a policy perspective, the results imply that, when well designed, carbon taxes can achieve environmental objectives without compromising price stability or consumer welfare in emerging-market economies.

The rest of the paper is structured as follows. Section 2 provides a brief background on South Africa's carbon tax. Section 3 describes the empirical strategy and data, while Section 4 presents the empirical results. Section 5 reports the results of robustness tests. Section 6 concludes.

2 Background on South Africa's Carbon Tax

South Africa is the largest emitter of GHG emissions in Africa, primarily driven by its coal-powered energy sector (see Figure 1). In June 2019,² South Africa became the first African country to implement a carbon tax, aligning with its ambitious climate objectives to reduce GHG emissions to between 398 and 510 MtCO₂e by 2025, 350 to 420 MtCO₂e by 2030, and to achieve net-zero emissions by 2050.³ The carbon tax operates under the *polluter-pays* principle, ensuring that economic agents (both firms and consumers) internalize the social cost of carbon when making production, consumption, and investment decisions. The overarching goal is to incentivize a transition toward low-carbon technologies and energy sources.

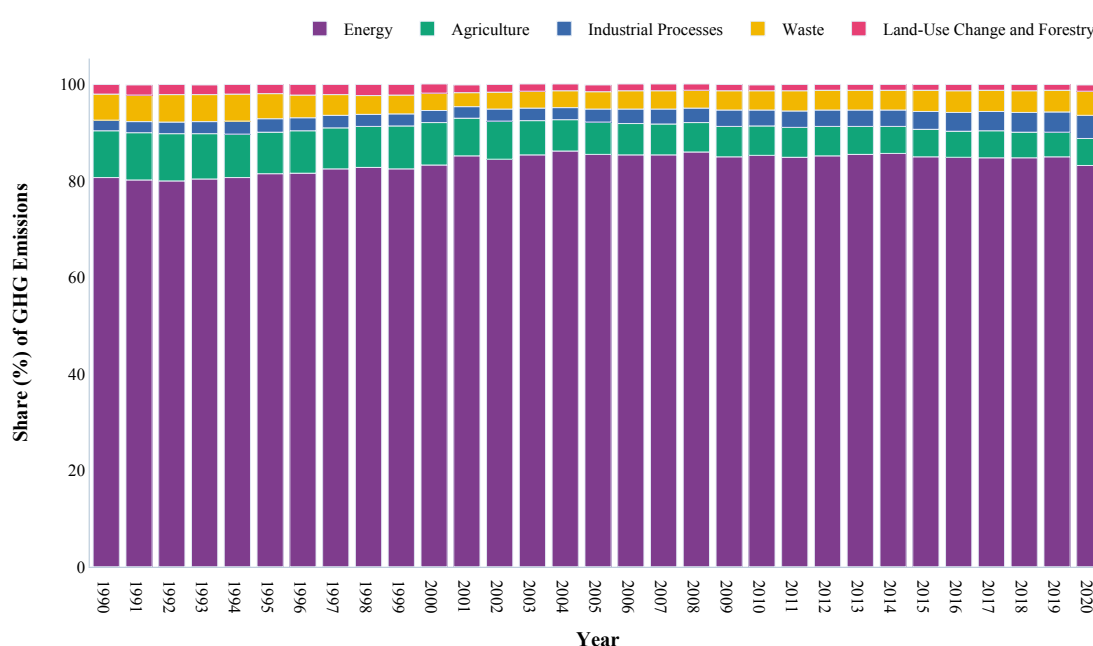


Figure 1: Sectoral Composition of South Africa's Historical GHG Emissions

Notes:

The implementation of the carbon tax is being carried out in phases. The first phase runs from June 1, 2019, to December 31, 2025,⁴ and the second phase is ex-

²The 2019 Carbon Tax Act No. 15 was signed into law in May 2019.

³<https://unfccc.int/sites/default/files/NDC/2022-06/South%20Africa%20updated%20first%20NDC%20September%202021.pdf>

⁴The first phase was initially scheduled to end on December 31, 2022, but was extended in the 2022

pected to cover the period from January , 2026, to December 31, 2030. The carbon tax currently applies to major GHG-emitting sectors, including fuel combustion, industrial processes, and fugitive emissions, while emissions from agriculture, forestry, land use, and waste are excluded from coverage.

At the time of enactment, the carbon tax rate was set at R120 (approximately US\$6.30) per tonne of CO₂e. According to the Carbon Tax Act, this rate increases annually by the rate of consumer price inflation plus two percentage points for the period up to December 31, 2022, as determined by Statistics South Africa. Thereafter, the rate is adjusted annually in line with consumer price inflation for the preceding tax year. Consequently, the carbon tax rate has risen over time and stands at R190 (approximately US\$10) per tonne of CO₂e as of January 2024.⁵

Budget Review.

⁵<https://www.treasury.gov.za/documents/National%20Budget/2024/review/Chapter%204.pdf>

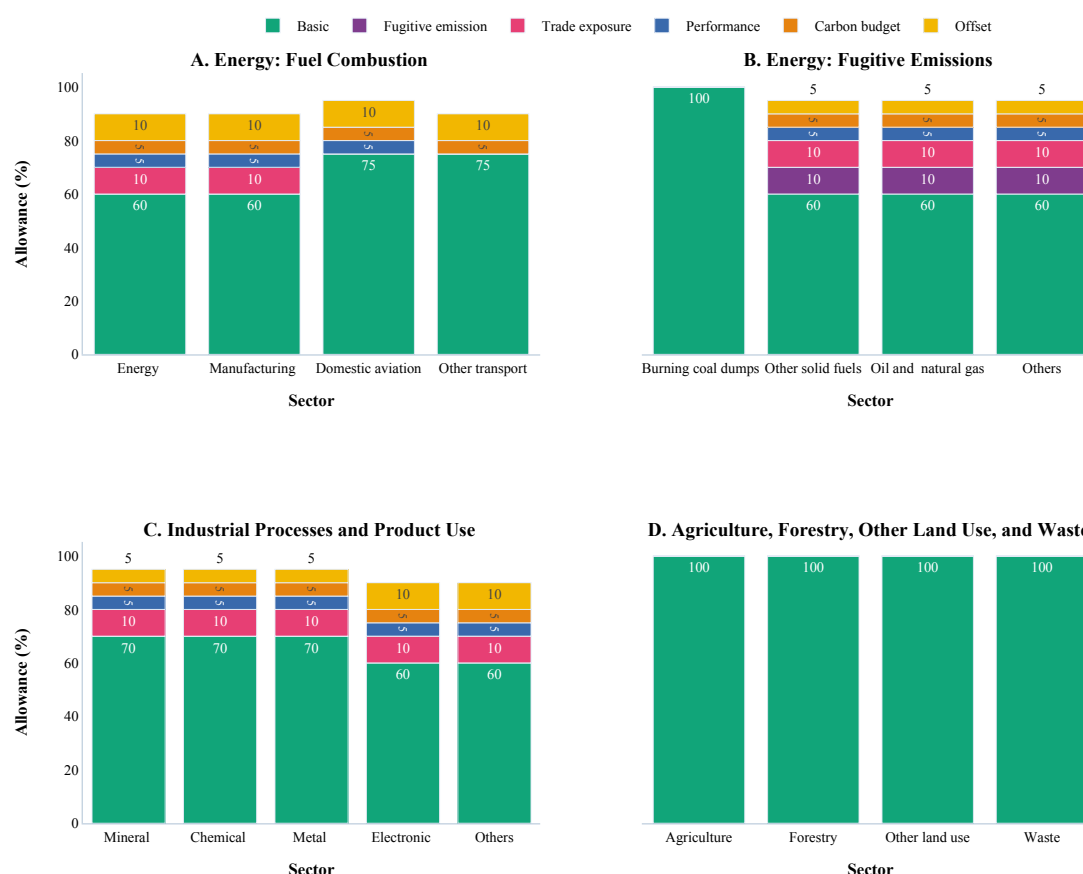


Figure 2: Sectoral Composition of South Africa's Historical GHG Emissions

Source: South Africa Carbon Tax Act, 2019

The Carbon Tax Act makes substantial provision for tax-free allowances during the initial phase, ranging from 60% to 95% (see Figure 2). These include a basic tax-free allowance of 60%–75% (depending on the sector), an additional 10% allowance for trade-exposed sectors, up to 5% for firms that reduce their emissions intensity, up to 5% for the use of carbon offsets, and up to 5% for companies complying with carbon-budget reporting requirements. These generous provisions mean that the effective carbon tax rate is substantially lower than the statutory rate. Moreover, Eskom—the national electricity utility and the country's largest single emitter—has been exempted from paying the carbon tax during the first phase, implying limited direct impact on electricity prices in the short term.

It is also important to note that South Africa's current carbon tax rate of R190

(about US\$10) per tonne of CO₂e remains below the estimated range of US\$40–80 per tonne required by 2030 to be consistent with the Paris Agreement temperature goals. Nonetheless, the government has proposed a gradual reduction of the carbon tax’s basic tax-free allowance from January 1, 2026, to December 31, 2030, in order to strengthen the price signal and enhance the tax’s effectiveness over time.⁶

3 Data and Empirical Strategy

3.1 Data

3.1.1 Consumer Prices

The outcome variables in this study are the headline Consumer Price Index (CPI), core CPI, energy CPI, and food CPI. Quarterly data for these series are obtained from the World Bank’s Global Inflation Database (Ha et al., 2023).

The use of quarterly data serves two main purposes. First, it allows for a sufficiently long pre-treatment period, conditional on data availability. A longer pre-treatment window enhances the credibility of the synthetic control by improving its ability to reproduce South Africa’s CPI trajectory prior to the introduction of the carbon tax. Second, quarterly data increase the likelihood of detecting short-lived or transient effects that may be obscured in annual data.

The selection of countries for the donor pool (excluding South Africa) is based on data availability and policy status. Specifically, we restrict the donor pool to developing countries with complete data for all four CPI components and that did not implement a carbon tax or emissions trading scheme (ETS) during the study period. This results in a balanced sample of 13 countries: Nigeria, Mauritius, Paraguay, Indonesia, the Philippines, Lao PDR, El Salvador, Malaysia, North Macedonia, Honduras, Guatemala, the Dominican Republic, and India. The sample period spans from 2015Q1 to 2022Q3.

⁶<https://www.treasury.gov.za/documents/National%20Budget/2024/review/Chapter%204.pdf>

For robustness analysis, we also use annual headline CPI data for Sub-Saharan African countries obtained from [World Bank \(2024\)](#).

3.1.2 Other Data

Following the existing literature on the drivers of inflation in developing countries ([Nguyen et al., 2017](#); [Alper et al., 2017](#); [Ha et al., 2019a,b](#); [Nachega et al., 2024](#)), we include additional covariates to account for macroeconomic dynamics that may influence consumer prices. Specifically, we control for quarterly real GDP growth and the quarterly local currency–US dollar exchange rate. Both series are sourced from the [IMF \(2024\)](#).

In the robustness analysis, we further employ annual data on trade balance and broad money supply, sourced from [World Bank \(2024\)](#). The definitions and sources of all variables used in the analysis are summarized in [Table A.1](#), while the descriptive statistics are presented in [Table A.2](#).

3.2 Empirical strategy

This paper employs the Generalized Synthetic Control Method (GSCM) proposed by [Xu \(2017\)](#) to assess the causal impact of South Africa’s carbon tax on consumer prices. The GSCM provides a data-driven approach to constructing a synthetic counterfactual using information from a pool of comparable countries that were not exposed to the intervention, following a framework similar to the conventional Synthetic Control Method (SCM) ([Abadie and Gardeazabal, 2003](#); [Abadie et al., 2010](#)).

The GSCM offers several advantages over the traditional SCM in this context. First, it allows for the inclusion of both country and time fixed effects, while also controlling for other observed and unobserved factors that may influence consumer prices. In particular, the incorporation of time fixed effects enables the model to flexibly account for global shocks—such as the COVID-19 pandemic and the Russia–Ukraine conflict—that could have affected inflation dynamics across countries. Second, the GSCM permits classical statistical inference by providing uncertainty estimates such

as standard errors and confidence intervals, unlike the conventional SCM, which relies on placebo-based inference.

The GSCM also improves upon the widely used difference-in-differences (DiD) framework. It relaxes the parallel trends assumption by modelling unobservable time-varying heterogeneity using an interactive fixed-effects (IFE) structure (Bai, 2009). This feature is particularly useful in cross-country settings where pre-treatment trends often diverge due to differences in macroeconomic structures, trade exposure, and policy environments.

3.2.1 GSCM Framework

Following Xu (2017), the outcome of country j at time (quarter) t can be expressed as:

$$Y_{jt} = \alpha_0 + \delta_{jt}carbontax_{jt} + \beta X'_{jt} + \lambda'_{jt}f_{jt} + \epsilon_{jt} \quad (3.1)$$

where Y_{jt} denotes the outcome of interest (consumer price index) for country j at time t ; $carbontax_{jt}$ equals 1 if country j is South Africa and $t \geq 2019Q2$, and 0 otherwise; X_{jt} is a $(k \times 1)$ vector of observed covariates; and β is the corresponding coefficient vector. The term f_t is an $(r \times 1)$ vector of unobserved common factors that capture time-varying global shocks, while λ_{jt} is an $(r \times 1)$ vector of country-specific factor loadings representing heterogeneous exposure to those shocks. ϵ_{jt} is the idiosyncratic error term.

The parameter of interest is δ_{jt} , which captures the causal effect of the carbon tax on consumer prices. Let Y_{jt}^1 and Y_{jt}^0 denote the potential outcomes for country j at time t under treatment and non-treatment, respectively. The treatment effect on the treated can thus be defined as:

$$\delta_{jt} = Y_{jt}^1 - Y_{jt}^0, \quad t > T_0 \quad (3.2)$$

3.2.2 Formulation of the Counterfactual

The main empirical challenge lies in the fact that Y_{jt}^0 —the counterfactual outcome for South Africa after the introduction of the carbon tax—is unobservable. To estimate it, a synthetic control is constructed from a pool of developing countries that did not implement carbon pricing during the study period. Let N_{tr} and N_{co} denote the number of treated and comparison units, respectively. For each country in the donor pool, the outcome equation can be written as:

$$Y_{jt} = \beta X'_{jt} + \lambda_{jt}' f_t + \epsilon_{jt} \quad (3.3)$$

Aggregating across all donor countries yields:

$$Y_{co} = \beta X'_{co} + F \Lambda'_{co} + \epsilon_{co} \quad (3.4)$$

where Y_{co} and ϵ_{co} are $(T \times N_{co})$ matrices, X_{co} is a $(T \times N_{co} \times k)$ matrix of covariates, and Λ_{co} is an $(N_{co} \times r)$ matrix of factor loadings. The identification of β , F , and Λ_{co} requires that the factors be normalized and orthogonal.

The counterfactual outcome, $\hat{Y}_{jt}^0$, is constructed in three steps:

1. **Estimation of factors and parameters:** Estimate $\hat{\beta}$, $\hat{F}$, and $\hat{\Lambda}_{co}$ using an interactive fixed-effects model (Bai, 2009):

$$(\hat{\beta}, \hat{F}, \hat{\Lambda}_{co}) = \underset{\tilde{\beta}, \tilde{F}, \tilde{\Lambda}_{co}}{\operatorname{argmin}} \sum_{j \in C} (Y_j - \tilde{\beta} X_j - \tilde{F} \tilde{\Lambda}_j)' (Y_j - \tilde{\beta} X_j - \tilde{F} \tilde{\Lambda}_j) \quad (3.5)$$

$$\text{s.t. } \frac{\tilde{F}' \tilde{F}}{T} = I_r \quad \text{and} \quad \Lambda_{co}' \Lambda_{co} = I_r$$

2. **Estimation of factor loadings for the treated unit:** For the treated country, estimate $\hat{\lambda}_j$ by minimizing the mean squared prediction error (MSPE) in the

pre-treatment period:

$$\begin{aligned}\hat{\lambda}_j &= \underset{\lambda_j}{\operatorname{argmin}} \sum_{j \in C} (Y_{0j} - \hat{\beta}X_{0j} - \hat{F}_0\lambda_j)' (Y_{0j} - \hat{\beta}X_{0j} - \hat{F}_0\lambda_j) \\ &= (\hat{F}_0'\hat{F}_0)^{-1}\hat{F}_0'(Y_{0j} - \hat{\beta}X_{0j}), \quad j \in C \quad (3.6)\end{aligned}$$

3. **Construction of the counterfactual:** Use the estimated parameters to predict the counterfactual outcome for the treated unit:

$$Y_{jt}^0 = \hat{\beta}x_{jt}' + \hat{\lambda}_{jt}'\hat{f}_t \quad (3.7)$$

Finally, the estimated average treatment effect on the treated (ATT) is computed as:

$$ATT_t = \frac{1}{N_{tr}} \sum_{j \in C} (Y_{jt}^1 - Y_{jt}^0), \quad t > T_0 \quad (3.8)$$

This framework enables the estimation of dynamic treatment effects while accounting for both observed covariates and unobserved heterogeneity across countries and over time.

4 Results

4.1 Time-Varying Impact of South Africa's Carbon Tax on Headline Consumer Prices

Figure 3 presents the time-varying effects of South Africa's carbon tax on headline CPI using the GSCM estimates. Panel A plots the evolution of South Africa's actual headline CPI (solid red line) and the synthetic control (dashed black line), with the introduction of the carbon tax in June 2019 denoted by the vertical blue dashed line. Prior to implementation, the synthetic control closely replicates South Africa's CPI trajectory, indicating a strong pre-treatment fit and validating the quality of the synthetic counterfactual. Following the implementation, the two series remain nearly

identical for approximately three quarters, after which a small divergence appears, with South Africa's CPI slightly exceeding that of the synthetic control between the fourth and thirteenth post-treatment quarters.

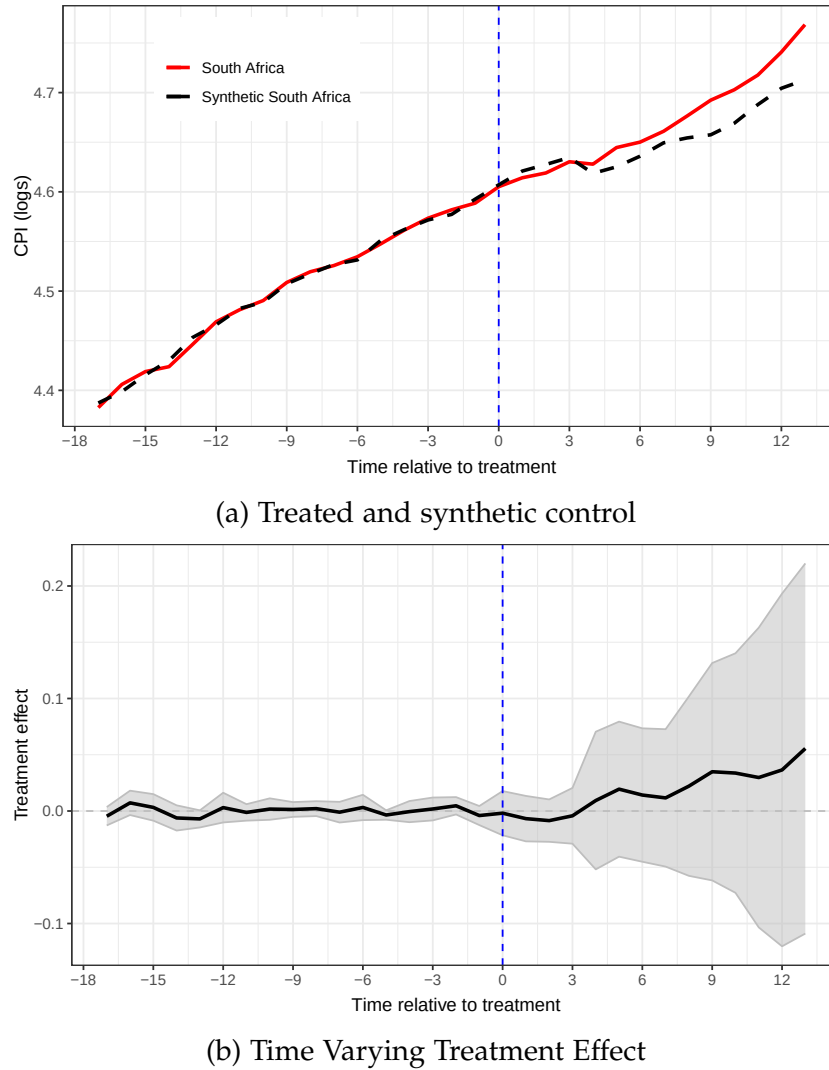


Figure 3: Impact of South Africa's Carbon Tax on Headline Consumer Prices

Notes: The figure presents the estimated effects of South Africa's carbon tax on the headline consumer price index (CPI) using the Generalized Synthetic Control Method (GSCM) proposed by Xu (2017). Panel (a) depicts the trajectories of South Africa's headline CPI and its synthetic control, while Panel (b) shows the corresponding time-varying treatment effects. The shaded gray area represents the 95% confidence interval around the estimated effects.

Panel B of Figure 3 plots the estimated treatment effects (the difference between the observed CPI and its synthetic counterpart). Since the dependent variable is expressed in logarithmic form, these differences can be interpreted as percentage deviations relative to the counterfactual. The shaded gray band denotes the 95% confidence

interval. Consistent with the visual evidence, the estimated effects remain statistically indistinguishable from zero across the entire post-treatment period. This implies that the introduction of the carbon tax did not exert a measurable effect on aggregate consumer prices.

These findings are consistent with evidence from advanced economies, such as Konradt et al. (2024), who document no inflationary impact of carbon pricing in the Euro Area. The absence of statistically significant effects on headline CPI aligns with the view that carbon pricing primarily shifts *relative* prices, with limited aggregate inflationary consequences (Konradt and Weder di Mauro, 2023; Konradt et al., 2024; Moessner, 2025). Quantitatively, the small post-treatment deviations in Panel B are well within the confidence bands typically found in cross-country carbon-pricing studies using panel and local-projection approaches (Moessner, 2025). Compared with findings of modest but temporary headline effects reported in some European settings (Berthold et al., 2023), our results sit at the lower bound of inflationary responses.

Two mechanisms are consistent with this: (i) South Africa’s generous first-phase allowances (up to 95%) and a relatively low statutory rate (R120 rising to R190 per tCO₂e) materially reduce effective marginal cost shocks; and (ii) partial insulation of administered/regulated prices (notably electricity during Eskom’s exemption) dampens short-run pass-through to the consumer basket.

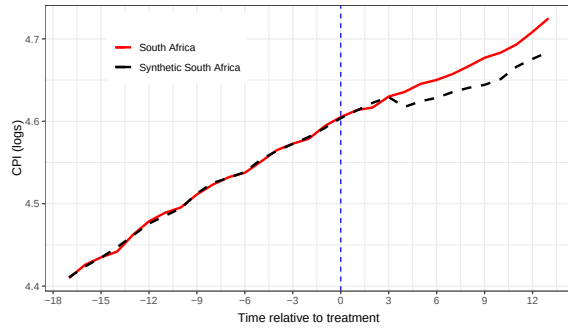
4.2 Time-Varying Impact on Disaggregated Consumer Price Indices

To explore potential heterogeneity across CPI components, we extend the analysis to core, energy, and food price indices. Carbon taxes are expected to have direct effects on energy prices, with potential indirect spillovers to food prices through higher production and transportation costs, whereas core inflation—which excludes energy and food—should remain largely unaffected (Baumeister et al., 2010; Gao et al., 2014).

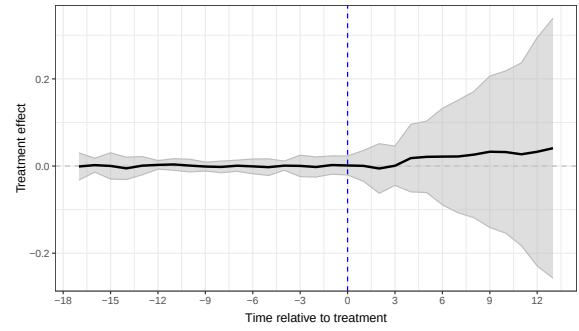
Figure 4 displays the estimated dynamic effects for each component. The synthetic controls closely replicate South Africa’s pre-treatment trajectories across all three in-

dices, confirming the reliability of the constructed counterfactuals. Post-treatment, the trends for core CPI and food CPI closely mirror the headline CPI pattern: deviations from the counterfactual are small and statistically insignificant.

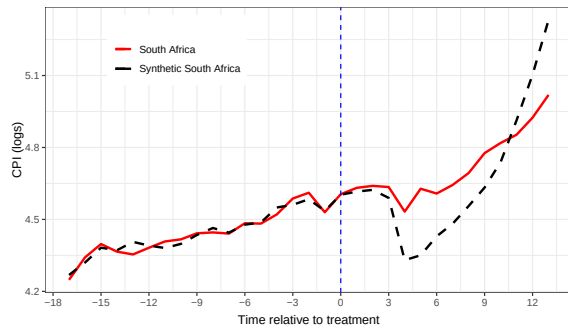
By contrast, the energy CPI exhibits a modest but more noticeable deviation. South Africa's energy CPI exceeds that of the synthetic control between the third and tenth quarters following the tax implementation, before converging again thereafter. Despite these short-lived differences, none are statistically significant at conventional confidence levels. These results suggest that while the energy component may have responded more directly to the carbon tax, the magnitude of the effect remains economically minor and transient.



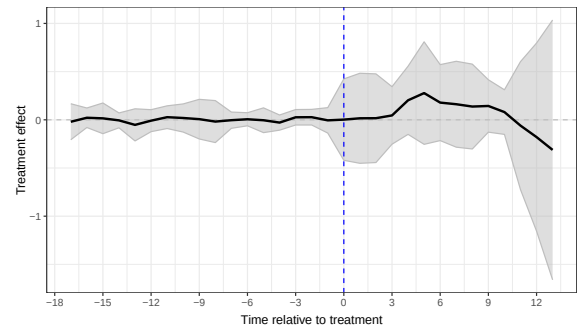
(a) Core CPI: Treated and Synthetic Control



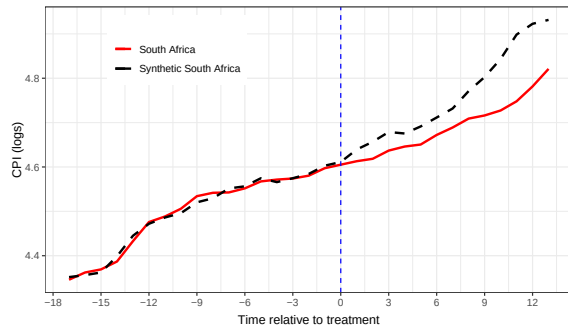
(b) Core CPI: Time Varying Treatment Effect



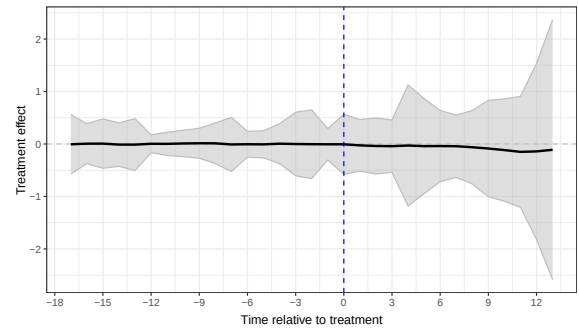
(c) Energy CPI: Treated and Synthetic Control



(d) Energy CPI: Time Varying Treatment Effect



(e) Food CPI: Treated and Synthetic Control



(f) Food CPI: Time Varying Treatment Effect

Figure 4: Impact of South Africa's Carbon Tax on Disaggregated Consumer Prices

Notes: The figure presents the estimated effects of South Africa's carbon tax on core, energy, and food consumer price indices, using the Generalized Synthetic Control Method proposed by Xu (2017). The shaded gray area represents the 95% confidence interval around the estimated treatment effects.

The component-level pattern matches theoretical priors and international evidence: energy shows the clearest (albeit modest) response, while core and food remain largely unaffected (Moessner, 2025). The muted food response suggests that indirect cost-push channels (energy-to-food) were either small, offset by other forces (e.g., exchange-rate appreciation cycles, supply-chain normalization, or competitive pricing

pressures), or diluted by firms' mark-up adjustments. In European markets, indirect spillovers to food and core have been found to be limited or short-lived (Konradt et al., 2024); our estimates concur, but for a developing-country setting where a priori energy intensity is higher. The divergence of energy CPI in early post-treatment quarters is consistent with partial direct pass-through, yet the absence of statistical significance points to the cushioning role of allowances, offsets, and phased design. The lack of core effects also accords with inflation-driver evidence in EMDEs where exchange rates, food supply shocks, and monetary conditions dominate core dynamics (Ha et al., 2019a,b).

4.3 Average and Cumulative Effects of the Carbon Tax on Consumer Prices

To complement the time-varying estimates, we compute both average and cumulative treatment effects of the carbon tax over the post-implementation period. This approach provides a broader perspective on whether the tax produced any sustained or aggregated price effects.

Table 1 reports the average treatment effects. In line with the dynamic analysis, the estimated coefficients for headline, core, and energy CPI are positive but statistically insignificant, indicating that, on average, the carbon tax has not raised consumer prices relative to the counterfactual.

Figure 5 illustrates the cumulative effects up to thirteen quarters after the policy's introduction. The cumulative responses of headline, core, and food CPI remain near zero and within the 95% confidence bounds, with standard errors increasing gradually over time. By contrast, the cumulative effect on energy CPI becomes positive from the third quarter, peaks at approximately 1.2% in the eleventh quarter, and stabilizes at roughly 1% after thirteen quarters. Although statistically significant at intermediate horizons, this effect is quantitatively small.

Table 1: GSCM Estimates of the Average Impact of South Africa’s Carbon Tax on Consumer Prices

	Headline	Core	Energy	Food
carbon tax	0.017 (0.030)	0.019 (0.064)	0.050 (0.036)	-0.066 (0.239)
Country FE	✓	✓	✓	✓
Year-Quarter FE	✓	✓	✓	✓

Notes: The table reports the Generalized Synthetic Control Method estimates of the impact of South Africa’s carbon tax on consumer prices. Robust standard errors, clustered at the country level, are in parentheses.

Taken together, these findings indicate that the carbon tax’s aggregate impact on consumer prices is negligible, with only a limited and modest adjustment in energy prices. This pattern supports the interpretation that carbon taxation primarily affects relative prices rather than overall inflation (Konradt and Weder di Mauro, 2023; Konradt et al., 2024). In the South African context, the muted effects likely reflect the tax’s design—particularly the generous emissions allowances during the first phase, which effectively reduce the tax burden for emitters and provide adjustment time. Moreover, Eskom, the state-owned electricity producer and largest emitter, was exempted from the tax during this phase, further dampening potential pass-through effects to consumer prices.

The cumulative energy effect of roughly 1% after thirteen quarters is economically small compared with stylized estimates from advanced-economy panels where a US\$10/tCO₂ increase is associated with about 0.8 pp in energy CPI and only 0.08 pp in headline inflation (Moessner, 2025). While cross-study comparability is imperfect (different samples, instruments, and identification designs), the order of magnitude reinforces the interpretation that South Africa’s first-phase carbon price signal was attenuated at the consumer level. This attenuation plausibly reflects: (i) large statutory allowances (60–95%), (ii) offset provisions and performance credits, (iii) Eskom’s exemption in phase 1, and (iv) policy credibility that may have reduced precautionary price-setting responses. That said, the upward cumulative profile on energy CPI from quarter 3 to quarter 11 is consistent with lagged adjustment along energy supply

chains (procurement cycles, contracts, and regulated tariff calendars) documented in energy pass-through studies (Baumeister et al., 2010). The lack of cumulative effects on headline, core, and food indicates limited propagation through broader cost networks, in line with the “relative-price reallocation” rather than “generalized inflation” view of carbon pricing (Konradt and Weder di Mauro, 2023; Konradt et al., 2024).

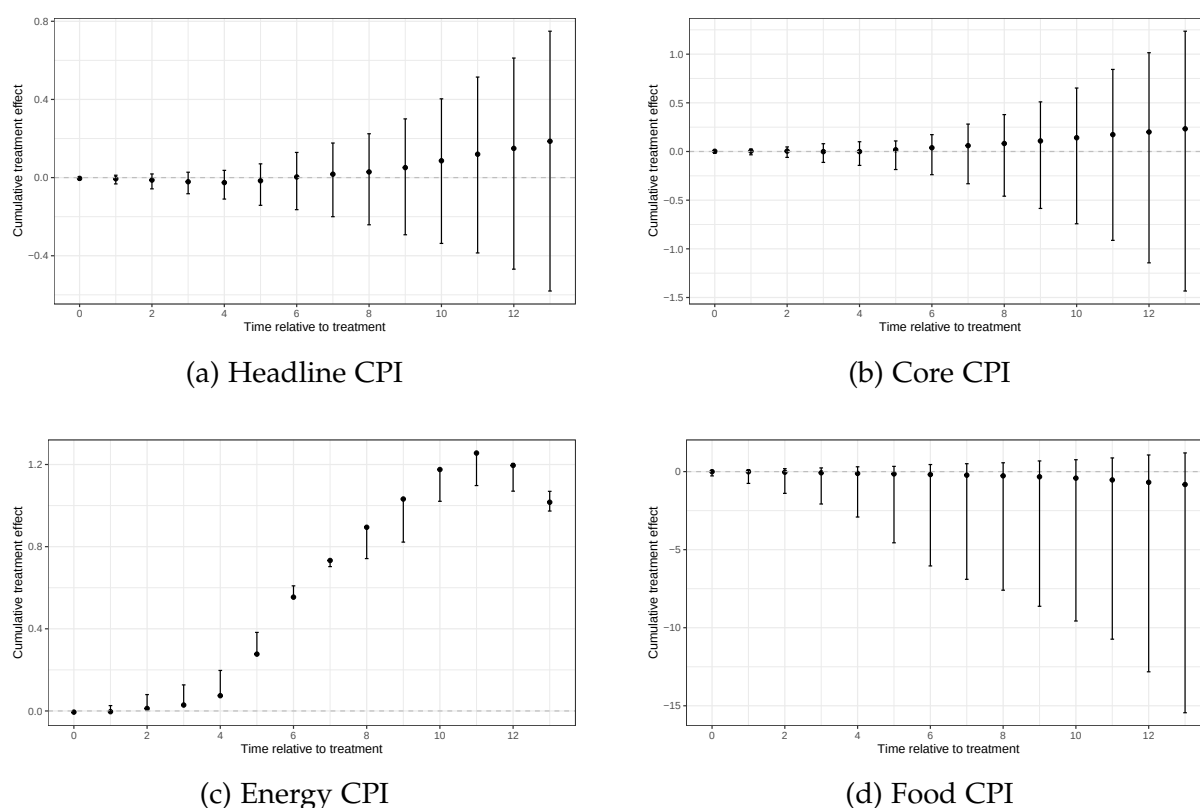


Figure 5: Cumulative Impact of South Africa's Carbon Tax on Consumer Prices

Notes: The figure presents the cumulative effects of South Africa's carbon tax on headline, core, energy, and food consumer price indices, estimated using the Generalized Synthetic Control Method proposed by Xu (2017). The horizontal bars represent the 95% confidence intervals around the estimated cumulative effects.

5 Robustness Checks

We conduct two sets of robustness checks to verify the stability of the baseline results.

5.1 Alternative Estimator

As a first check, we re-estimate the average treatment effects using the Synthetic Difference-in-Differences (SDiD) method recently developed by [Arkhangelsky et al. \(2021\)](#). The SDiD estimator combines features of the conventional SCM and the traditional difference-in-differences approach. Like the SCM, it assigns data-driven weights to control units based on their similarity to the treated unit’s pre-treatment trajectory, while also permitting classical inference similar to the GSCM.

A key distinction is that SDiD adjusts the weights to ensure approximate parallel trends between the treated and synthetic control units, improving comparability and robustness to specification. It also emphasizes time periods that are more similar to the treated period, thereby strengthening internal validity.

[Table A.3](#) reports the SDiD estimates of the average treatment effect. The results are virtually identical to those obtained from the GSCM, confirming that the estimated effects are robust to the choice of estimator.

Concordant SDiD and GSCM estimates mitigate concerns that our findings hinge on one estimator’s functional-form or weighting scheme. Given SDiD’s explicit targeting of parallel trends and GSCM’s capacity to absorb latent common factors, convergence across methods strengthens internal validity. Similar triangulation strategies are increasingly recommended in policy-evaluation studies of carbon pricing, precisely because shocks of interest are small relative to macro volatility.

5.2 Restricting the Donor Pool to Sub-Saharan Africa

As a second robustness check, we restrict the donor pool to Sub-Saharan African (SSA) countries and conduct the analysis using annual headline CPI data, given the limited availability of high-frequency data for most African economies. The analysis spans

2000–2022, with 2019 as the treatment year. The donor pool includes Nigeria, Mauritius, Ghana, Burkina Faso, Cape Verde, Uganda, Zimbabwe, Kenya, and Madagascar. In addition to real GDP growth and the exchange rate, we control for trade balance and broad money supply.

Panel A of [Figure A.1](#) plots the evolution of South Africa’s annual headline CPI and its synthetic counterpart. The fit of the synthetic control in this specification is somewhat less precise than in the quarterly analysis, reflecting the coarser data frequency. Nevertheless, Panel B shows that the difference between South Africa’s CPI and that of the synthetic control remains statistically indistinguishable from zero in both pre- and post-treatment periods, and Panel C confirms no evidence of a cumulative effect.

Limiting the donor pool to SSA improves contextual comparability at the cost of sample size and pre-treatment fit (a standard trade-off in synthetic-control applications). The persistence of null headline effects in this setting indicates that the baseline conclusions are not driven by cross-regional composition of the donor pool. Moreover, annual-frequency results mirror the quarterly patterns for aggregate CPI, suggesting that any short-lived dynamics missed by annual data are not large enough to alter medium-run conclusions.

6 Conclusion and Policy Implications

Countries are increasingly implementing carbon pricing instruments—such as carbon taxes and emissions trading schemes to internalize the social costs associated with GHG emissions and to incentivize a transition toward cleaner energy sources. Despite their environmental rationale, concerns persist that carbon pricing may exacerbate inflationary pressures. In theory, carbon pricing can influence inflation through both supply- and demand-side channels: it may raise production costs and consumer prices in energy-intensive sectors, but it can also dampen aggregate demand and investment by reducing disposable income. The net effect is therefore ambiguous and

context-dependent, making the inflationary impact of carbon pricing an empirical question. Yet, the existing empirical literature largely focuses on advanced economies, leaving limited evidence from developing countries where energy intensity and policy design differ substantially.

This paper fills this gap by assessing the causal impact of South Africa’s carbon tax on consumer prices using a quasi-experimental approach based on the Generalized Synthetic Control Method . Our analysis yields three main findings. First, we find no statistically significant impact of the carbon tax on headline consumer prices relative to the counterfactual, suggesting that the tax has not been inflationary on aggregate. Second, we detect a small but positive cumulative effect—peaking at roughly 1%—on the energy component of the CPI, while effects on core and food prices remain negligible. Third, these results are robust to alternative estimation using the Synthetic Difference-in-Differences method and to restricting the donor pool to Sub-Saharan African countries.

These findings support the view that carbon taxation primarily induces *relative price adjustments* rather than broad-based inflationary pressures (Konradt and Weder di Mauro, 2023; Konradt et al., 2024). In South Africa’s case, the limited aggregate impact likely reflects key features of the current tax design: generous emissions allowances of up to 95% during the first phase, the phased rollout of the tax, and the exemption of Eskom—the country’s largest emitter and electricity producer—from paying the tax in the initial implementation period. These provisions substantially reduce effective cost pass-through to final consumers, while offering firms time to adjust production processes and invest in cleaner technologies.

From a policy perspective, the results offer several insights. First, they suggest that well-designed carbon taxes can achieve environmental objectives without undermining price stability, alleviating fears of “greenflation.” Second, the muted pass-through observed in South Africa underscores the importance of gradual implementation and transitional support mechanisms in maintaining both economic and political feasibility, particularly in developing economies. Finally, as South Africa enters the second

phase of its carbon tax—during which free allowances are expected to be gradually reduced—continued monitoring will be essential to evaluate whether stronger price signals alter inflation dynamics or improve emissions abatement. In this regard, future research could examine distributional effects, firm-level price-setting responses, and the interaction between carbon pricing and monetary policy in emerging markets.

While this study provides an ex-post causal evidence on the inflationary impact of a carbon tax in a developing-country context, several limitations warrant further exploration. The analysis focuses on aggregate price indices and therefore cannot capture heterogeneity across income groups or sectors. Future research could extend the framework to micro-level or firm-level data to explore distributional impacts, cost pass-through mechanisms, and price-setting behavior. Moreover, integrating the results into a general equilibrium or dynamic macroeconomic model could shed light on longer-term interactions between carbon pricing, fiscal adjustments, and monetary policy transmission in emerging-market settings.

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Appendix A: Additional Tables and Figures

Table A.1: Variable Definitions and Data Sources

Variable	Frequency	Definition	Source
Headline CPI	Quarterly and Annual	Cost to the average consumer of acquiring a basket of goods and services index	Ha et al. (2023); World Bank (2024)
Core CPI	Quarterly	Cost to the average consumer of acquiring a basket of goods and services index	Ha et al. (2023)
Energy CPI	Quarterly	Cost to the average consumer of acquiring energy services	Ha et al. (2023)
Food CPI	Quarterly	Cost to the average consumer of acquiring a basket of food and non-alcoholic beverages	Ha et al. (2023)
Exchange Rate	Quarterly and Annual	National currency per U.S. dollar (period average)	IMF (2024)
GDP	Quarterly and Annual	Real gross domestic product, seasonally adjusted	IMF (2024)
Trade Balance	Annual	Net trade in goods and services	World Bank (2024)
Broad Money	Annual	Percent of GDP	World Bank (2024)

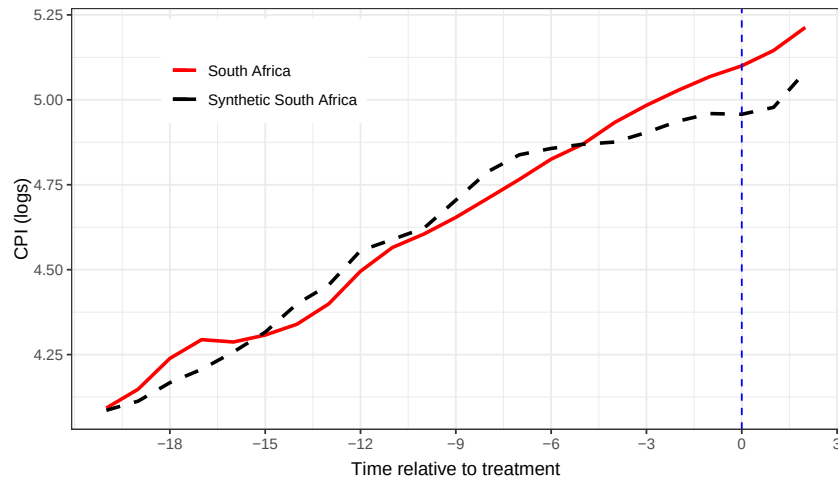
Table A.2: Summary Statistics

	N	Mean	Min	Max
Headline consumer price index	496	135.61	97.00	432.59
Core consumer price index	496	140.41	88.13	395.97
Energy price index	496	130.74	1.20	425.13
food price index	496	151.11	82.22	551.24
Exchange rate	496	1502.01	1.00	14989.86
Real GDP	248	3.56e+08	44735.30	2.93e+09

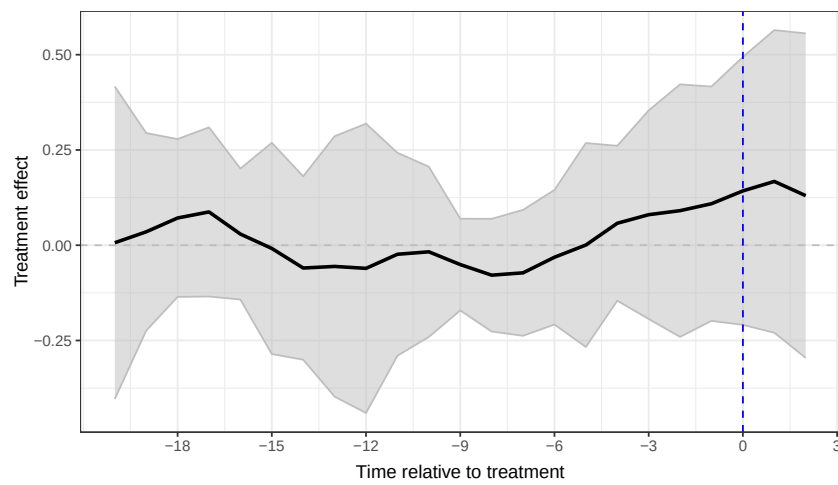
Table A.3: SDiD Estimates of the Average Impact of South Africa's Carbon Tax on Consumer Prices

	Headline	Core	Energy	Food
carbon tax	0.000 (0.023)	-0.005 (0.021)	0.150 (0.121)	-0.043 (0.044)
Observations	248	248	248	248

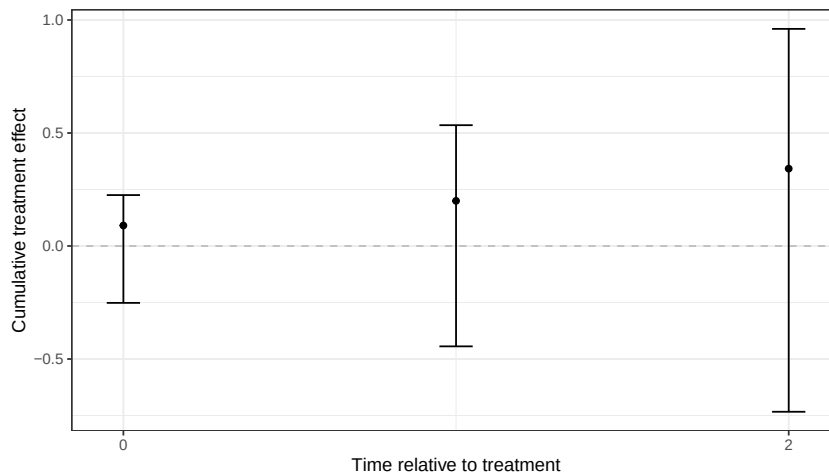
Notes: The table reports the Synthetic Difference-in-Differences estimates of the impact of South Africa's carbon tax on consumer prices. Control variables include real GDP growth and the exchange rate.



(a) Treated and synthetic control



(b) Time Varying Treatment Effect



(c) Cumulative Treatment Effect

Figure A.1: Impact of South Africa's Carbon Tax on Headline Consumer Prices Using Annual Data

Notes: The figure presents the estimated effects of South Africa's carbon tax on the annual headline consumer price index, using the Generalized Synthetic Control Method proposed by Xu (2017). Panel (a) depicts the trajectories of South Africa's headline CPI and its synthetic control; Panel (b) shows the corresponding time-varying treatment effects; and Panel (c) illustrates the cumulative effects. The shaded gray areas and horizontal bars represent the 95% confidence intervals around the estimated effects.